

EDUCATION	Purdue University	
	<i>Ph.D. Candidate in Mathematics</i>	May 2027 (expected)
	• Research Interests: Inverse Problems & Partial Differential Equations • Advisors: Peijun Li & Isaac Harris	
	New Jersey Institute of Technology	
PUBLICATIONS & PREPRINTS	<i>M.S. in Applied Mathematics</i>	May 2021
	• Advisor: Christina Frederick	
	Rutgers University	
	<i>B.A. in Mathematics</i>	May 2018
WORK EXPERIENCE	1. R. Ayala, I. Harris, G. Ozochiawaeze . Factorization method for the biharmonic scattering problem for an absorbing penetrable obstacle. <i>Submitted</i> , 2025. arXiv:2511.05711	
	2. I. Harris, P. Li, G. Ozochiawaeze . Sampling methods for the inverse cavity scattering problem of biharmonic waves. <i>Inverse Problems</i> , 2025. DOI: 10.1088/1361-6420/ac2ef8	
	MIT Lincoln Laboratory Lexington, MA	May 2023 – Oct 2023
	• <i>Summer Research Intern, Group 36 – Integrated Missile Defense Technology</i> • Applied compressive sensing to enhance radar imaging for missile target detection, improving reconstruction accuracy. • Presented compelling research findings on compressed radar sensing, demonstrating potential applications in enhanced defense technologies.	
	MIT Lincoln Laboratory Lexington, MA	May 2022 – Aug 2022
	• <i>Summer Research Intern, Group 37 – Advanced Undersea Systems & Technology</i> • Developed algorithms to analyze coherence loss in sonar signals, improving underwater detection and tracking. • Applied sonar array interferometry to extract spatial information from acoustic wave-fronts.	
	New Jersey Institute of Technology Newark, NJ	Jul 2020 – May 2021
	• <i>Acoustic Simulation Researcher</i> • Devised oceanographic models to simulate acoustic wave propagation in underwater environments in COMSOL. • Critically applied finite element methods and domain decomposition modeling to simulate high-frequency acoustic wave behavior in complex ocean conditions.	
TEACHING	• Lecturer, Purdue University — MA 16020: <i>Applied Calculus II</i>, Fall 2025 • Recitation Instructor, Purdue University — MA 261: <i>Multivariable Calculus</i>, Spring 2024 • Teaching Assistant, Purdue University — MA 511: <i>Linear Algebra and Applications</i>, Spring 2023 • Recitation Instructor, Purdue University — MA 261: <i>Multivariable Calculus</i>, Fall 2022 • Recitation Instructor, Purdue University — MA 162: <i>Calculus II</i>, Spring 2022 • Recitation Instructor, Purdue University — MA 161: <i>Calculus I</i>, Fall 2021 • Instructor, Accel Learning — Mentored students in preparation for AMC and AIME math competitions, Spring–Fall 2020 • Math Instructor, Mathnasium, K-12 Teaching & Tutoring, Fall 2018–Spring 2019	

CONFERENCES & TALKS	SIAM Conference on Analysis of Partial Differential Equations (PD25), Pittsburgh (invited speaker)	2025
	<i>A Factorization Method Approach to the Biharmonic Transmission Problem in Absorbing Media</i>	
	SIAM Conference on Analysis of Partial Differential Equations (PD25), Pittsburgh (speaker)	2025
	<i>Localizing Cavities in Biharmonic Scattering with Limited Data</i>	
	CCAM Lunch Seminar, Purdue University (speaker)	2025
	<i>Localizing Cavities in Biharmonic Scattering with Limited Data</i>	
	Summer School: Theory and Applications of Elliptic PDE, UC Irvine (speaker)	2025
	<i>Extending Qualitative Reconstruction to Biharmonic Scattering with Limited Data</i>	
	Graduate Student Analysis Seminar, Purdue University (speaker)	2025
	<i>Extending Qualitative Reconstruction in Biharmonic Scattering with Limited Data</i>	
	CCAM Lunch Seminar, Purdue University (speaker)	2025
	<i>On Sampling Methods for Recovering a Clamped Cavity in a Thin Plate</i>	
	Mathematics and Statistics in Industry Workshop and Panel Discussion (attendee)	2025
	Conference on Mathematics of Wave Phenomena, Karlsruhe Institute of Technology (attendee)	2025
	Purdue Graduate Research Day, Purdue University (speaker)	2024
	<i>A Linear Sampling Method for Recovering a Clamped Cavity in a Thin Plate</i>	
	SIAM Student Chapter Conference, Purdue University (speaker)	2024
	<i>A Linear Sampling Method for Recovering a Clamped Cavity in a Thin Plate</i>	
	Paris-Saclay Conference in Analysis and PDE, Laboratoire de Mathématiques d'Orsay (attendee)	2024
	Graduate Student Analysis Seminar, Purdue University (speaker)	2023
	<i>Comparing and Integrating the Probe Method and Method of Singular Sources</i>	
	Graduate Student Analysis Seminar, Purdue University (speaker)	2023
	<i>The Factorization Method in Inverse Scattering</i>	
	Integrated Missile Defense Technology Group, MIT Lincoln Laboratory (speaker)	2023
	<i>Compressive Radar Imaging in Missile Defense</i>	
	Advanced Undersea Systems & Technology Group, MIT Lincoln Laboratory (speaker)	2022
	<i>Quantifying Real Operational Performances of SONAR Arrays in Random Ocean Environments</i>	
	Purdue Math Student Colloquium, Purdue University (speaker)	2022
	<i>Finite Element Modeling of Underwater Acoustic Environments</i>	
	Workshop on Scientific Computing, Purdue University (attendee)	2022
	New Ideas in Computational Inverse Problems, BIRS, Banff International Research Station (attendee)	2022

INFORMAL TALKS	The Dawn of the Diophantine Era: Algebra at the Crossroads of Empires <i>Math History Seminar, Purdue University</i>	2025
	Fixed Points in Action: from Brouwer to Lefschetz & Beyond <i>Topolodays, Purdue University</i>	2025
	The Life, Mathematics, and Political Advocacy of Sofya Kovalevskaya <i>Math History Seminar, Purdue University</i>	2024
	The K-theoretic Atiyah-Singer Index Theorem <i>Student Operator Algebras Seminar, Purdue University</i>	2024
	Introducing Pseudodifferential Operators <i>Purdue Math Student Colloquium</i>	2024
	Bourbaki and the Foundations of Modern Math <i>Math History Seminar, Purdue University</i>	2023
	Wasan: Mathematical Treasures in Early Modern Japan <i>Math History Seminar, Purdue University</i>	2022
AWARDS AND HONORS	• SIAM (Society of Industrial & Applied Mathematics) Student Travel Award	2025
	• NSF Computational Mathematics Program Award, DMS-2208256,	2023-2024
	• GEM Fellowship	2022
	• Graduate C/Startup Research Grant	2020
	• Tau Sigma Honors Society	2016
SKILLS	Computing & Software: Python, C++, MATLAB, COMSOL, FEniCS	
SERVICE	Purdue Math Department Graduate Representative <i>Purdue University</i>	2025-2026
	Mentor, Directed Reading Program in Computational Topology <i>Purdue University — mentee: Aaron (Sang Hyun) Kim</i>	2025
	Founder and Organizer, Math History Seminar <i>Purdue University</i>	2022
	Rutgers Undergraduate Mathematics Association Treasurer & Web Manager <i>Rutgers University</i>	2017-2018